Ice-Unet: A Very High-Resolution UAV-Optimized Deep Learning Model for Sea Ice Classification

Yu Zhang, Zijun Dong, Tefeike Saideaihemaiti, Ezhar Elijan, Feng Xiao, Shengkai Zhang, and Fei Li

***Abstract*—Polar sea ice monitoring is critical for global climate research and Arctic ship navigation, yet traditional satellite remote sensing struggles to resolve microscale ice features due to resolution and weather constraints. This study presents a novel sea ice classification method integrating uncrewed aerial vehicles (UAV)-derived high-resolution optical imagery from the 2024 "Sun Yat-sen University Polar Vessel" Arctic expedition with deep learning. The dataset, comprising four types (sea ice, melt ponds, open water, and ships), was constructed through geometric correction and semiautomated annotation using the ISAT-SAM approach. The proposed Ice-Unet model, optimized via transfer learning from pretrained weights, employs a VGG16-inspired backbone with multiscale feature fusion and downscaling mechanisms for semantic segmentation. Comparative evaluations against DeepLabV3+ (Xception/MobileNet), PSPNet (ResNet50), and HRNet demonstrate Ice-Unet's superiority, achieving a leading IoU of 95.07% and recall rates of 97.11%, particularly excelling in melt pond detection and ice lead morphology characterization. Case analysis reveals precise quantification of melt pond spatial patterns and predictive capability for their evolution into sea ice leads. The research empirically confirms the operational synergy between UAV platforms and deep learning architectures in monitoring transient sea ice dynamics, establishing a submeter resolution methodology that significantly enhances the precision of sea ice classification-based ship navigation through near-real-time geospatial analytics.**

***Index Terms*—Ice-Unet, melt ponds, sea ice classification, uncrewed aerial vehicles (UAV).**

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Yu Zhang, Feng Xiao, and Shengkai Zhang are with Chinese Antarctic Center of Surveying and Mapping, the Key Laboratory of Polar Environment Monitoring and Public Governance, Ministry of Education, and the School of Geodesy and Geomatics, Wuhan University, Wuhan 430079, China, and also with the Luojia Laboratory of Hubei Province, Wuhan 430072, China (e-mail: yuzhang_spl@whu.edu.cn; shaw89@whu.edu.cn; zskai@whu.edu.cn).

Zijun Dong is with the School of Electronic and Information, Wuhan University, Wuhan 430072, China (e-mail: 2022302191102@whu.edu.cn).

Tefeike Saideaihemaiti and Ezhar Elijan are with the School of Geodesy and Geomatics, Wuhan University, Wuhan 430072, China (e-mail: 2022302141274@whu.edu.cn; 2022302141280@whu.edu.cn).

Fei Li is with the Department of Earth and Space Sciences, Southern University of Science and Technology, Shenzhen 518055, China (e-mail: lif3@sustech.edu.cn).

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I. Introduction

As a critical regulator of Earth's climate system, sea ice modulates polar ecosystems and ocean circulation through ice–atmosphere–ocean interactions. Accurate sea ice classification underpins polar remote sensing by providing essential parameters for concentration/thickness retrieval and deformation analysis, crucial for Arctic navigation safety and climate monitoring [1], [2]. While satellite sensors (e.g., passive microwave) offer wide coverage, their resolution of over 1 km and cloud sensitivity limit microtopography detection (e.g., melt pond heterogeneity). Uncrewed aerial vehicles (UAV)-based systems in submeter resolution with multispectral-thermal synergy and deep learning fusion algorithms now address these fine-tune target extraction issues [3], [4]. The high-resolution UAV platforms can also be equipped with multiplatform synergy for simultaneous multiparameter measurements. For example, integrated satellite-UAV-ship networks in the Bohai Sea achieve small than 1.2-m spatial registration errors, reducing observation costs to 7.3% of helicopters and logistics by 83%. Besides, the deep learning-based models have significantly enhanced both observation accuracy and processing efficiency through automated feature extraction and large-scale data analysis. Mask R-CNN automates floe extraction (IoU 87%) with 15× efficiency gains. These innovations position UAVs as key nodes in polar sensor networks, synergizing with satellite remote sensing to quantify sea ice–atmosphere–ocean [5]. Besides, Transformer architectures and attention mechanisms such as SegFormer and Swin-Unet also exhibit robust performance in semantic segmentation results on large-scale benchmark datasets [6].

In the proposed framework of very high-resolution UAV-optimized deep learning model for Arctic sea ice classification, a three-component Ice-Unet model, including a backbone feature extraction network, an enhanced feature extraction network, and a prediction network, is constructed. This study analyzes 2024 Arctic optical data from the "Sun Yat-sen University Polar Research Vessel," applying geometric correction, radiometric calibration, and orthorectification to ensure geometric/spectral consistency. A deep learning framework optimizes five semantic segmentation models through multiscale feature fusion and attention mechanisms.

The structure of this article is detailed as follows. Section II illustrates the brief information of the study area and data. Section III develops algorithm optimization strategies. Section IV quantifies classification and melt pond estimation using the proposed algorithm. Section V draws the conclusion. Results confirm UAVs' capacity to resolve subme-

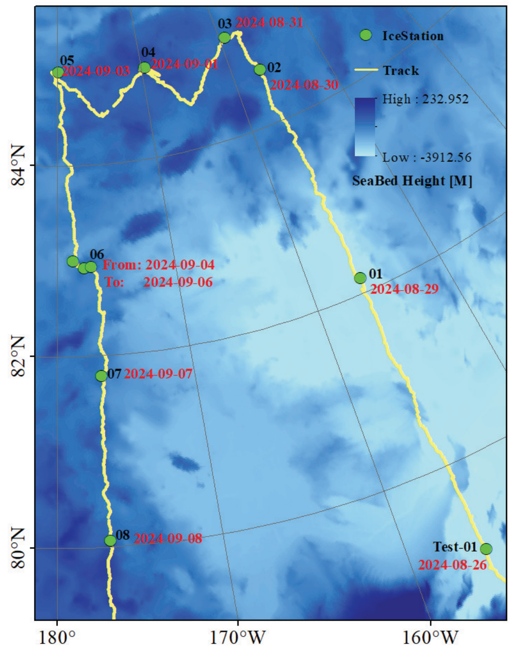

Fig. 1. Study area (base map is the ETOPO Seabed Height).

ter deformation features critical for sea ice mechanics while providing distribution of the melt ponds, advancing predictive capabilities for Arctic environmental change.

II. Study Area and Experiment Data

The UAV datasets primarily encompass four categories: optical orthophoto images, light detection and ranging (LiDAR), thermal infrared, and four-component sensor data. Among these, orthophoto images serve critical roles in sea ice observations and ship navigation. These images were acquired using UAV platforms (DJI Mavic 3 Enterprise and Feima V500), covering a total area of 13.05 km^2 with spatial resolutions ranging from 0.02 to 0.04 m, achieved at flight altitudes between 60 and 190 m. Data collection was conducted across multiple Arctic Ocean sampling sites (illustrated in Fig. 1). Notably, the process was influenced by sea ice drift at velocities of 0.2–0.6 km/h, necessitating real-time adjustments to the UAV return points to ensure positional accuracy and dataset integrity. We totally obtained 10 000 optical images in 11 ice stations (Fig. 1). The name (black) and the observation time (red) are marked on the map, and the difference of Ice Station 06 is caused by the sea ice drifting.

III. Ice-Unet for Sea Ice Classification

A. Geometric Correction and Radiation Calibration

Based on the ground reference point established by the observation data from in situ measurement, we constructed a precise parameter model between the image sensor and the ground reference point shown as follows:

$$\begin{cases} X = a_0 + a_1x + a_2y + a_3x^2 + a_4xy + a_5y^2 \\ Y = b_0 + b_1x + b_2y + b_3x^2 + b_4xy + b_5y^2 \end{cases} \tag{1}$$

where (X, Y) is the geographic coordinate, (x, y) is the image coordinate, and (a, b) is the polynomial coefficient. Then, coordinate transformation is carried out by using a second-order polynomial, and the resampling is carried out by using bilinear interpolation to obtain the corrected optical image. The coordinate transformation formula is as follows:

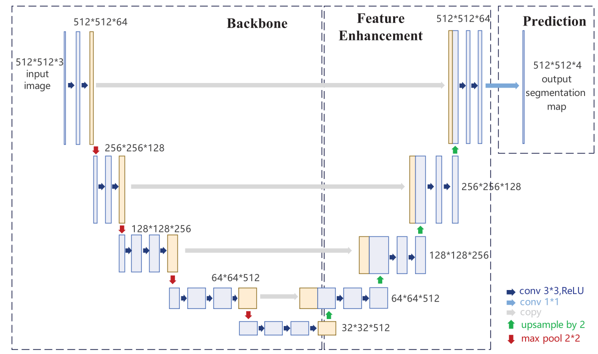

Fig. 2. Ice-Unet for sea ice classification.

$$\begin{cases} x' = c_0 + c_1X + c_2Y + c_3X^2 + c_4XY + c_5Y^2 \\ y' = d_0 + d_1X + d_2Y + d_3X^2 + d_4XY + d_5Y^2 \end{cases} \tag{2}$$

where x' and y' are the corrected image coordinates. On this basis, combined with the calibration coefficient of the sensor, including radiation gain (G) and offset (B), we convert the digital quantization value (DN value) into the radiation brightness value. The specific conversion formula is as follows:

$$L = G \cdot DN + B. \tag{3}$$

The final corrected radiation brightness will be obtained by considering L_{atm} and transmittance (τ)

$$L_{\text{ice}} = \frac{L - L_{\text{atm}}}{\tau}. \tag{4}$$

B. Orthorectification Stitching

For overlapping UAV imagery, SIFT extracts feature points matched across the adjacent images. Triangulation techniques are applied to accurately calculate the 3-D coordinates of these feature points as follows:

$$\begin{cases} X = \dfrac{(x_1 - x_0) \cdot Z}{f} \\ Y = \dfrac{(y_1 - y_0) \cdot Z}{f} \end{cases} \tag{5}$$

where X, Y, and Z are 3-D coordinates. Z represents the elevation value, acquired synchronously by the LiDAR sensor on the UAV. x_0 and y_0 are the image center, and x_1 and y_1 are the feature points. The collinear equation is established according to the UAV image for orthorectification, and the formula is as follows:

$$\begin{cases} x = -f \cdot \dfrac{a_1(X - X_0) + b_1(Y - Y_0) + c_1(Z - Z_0)}{a_3(X - X_0) + b_3(Y - Y_0) + c_3(Z - Z_0)} \\ y = -f \cdot \dfrac{a_2(X - X_0) + b_2(Y - Y_0) + c_2(Z - Z_0)}{a_3(X - X_0) + b_3(Y - Y_0) + c_3(Z - Z_0)}. \end{cases} \tag{6}$$

With the rotation matrix a_i, b_i, and c_i and the camera position $\mathrm{X_0}, \mathrm{Y_0}$, and $\mathrm{Z_0}$, the weighted average method is used to deal with the overlapping area as follows:

$$I_{\text{overlap}} = \omega_1 \cdot I_1 + \omega_2 \cdot I_2 \tag{7}$$

where I and ω are the gray value and the corresponding weight parameter, respectively. Final mosaic images via geographic

TABLE I
EVALUATION OF DIFFERENT METHODS

Metric	Network	Average	Sea Ice	Melt Pond	Sea Ice Leads	Ship
IoU/%	Ice-Unet	**95.07**	**96.47**	**91.46**	**94.86**	**97.50**
	DeepLabV3+(MobileNet)	86.87	93.67	85.01	52.80	0
	DeepLabV3+(Xception)	89.83	92.10	80.83	82.61	83.80
	PSPNet(ResNet50)	88.02	89.76	76.07	66.26	79.99
	HRNet	86.21	92.39	82.47	32.06	22.36
Recall/%	Ice-Unet	**97.11**	**98.29**	**95.35**	**96.06**	**98.74**
	DeepLabV3+(MobileNet)	87.24	97.11	91.13	89.63	0
	DeepLabV3+(Xception)	91.21	97.06	86.72	82.99	98.05
	PSPNet(ResNet50)	89.67	95.14	85.05	83.91	94.57
	HRNet	85.35	96.39	90.52	35.58	27.24
Precision/%	Ice-Unet	**97.44**	**98.11**	**95.73**	98.70	98.72
	DeepLabV3+(MobileNet)	95.02	96.36	92.68	56.24	0
	DeepLabV3+(Xception)	94.05	94.74	92.25	**99.45**	85.22
	PSPNet(ResNet50)	92.21	94.07	87.81	75.90	83.84
	HRNet	94.15	95.7	90.27	98.08	**100**

coordinates, optimized stitching lines, and multiresolution fusion to preserve spatial/spectral consistency, which provides a high-quality data basis for subsequent sea ice feature extraction and classification.

C. *Ice-Unet for Sea Ice Classification*

The proposed Ice-Unet framework, as illustrated in Fig. 2, comprises three core components: a backbone feature extraction network, an enhanced feature extraction network, and a prediction network. The structural details are elaborated as follows.

The backbone feature extraction network adopts a VGG16-inspired architecture, where the input sea ice image with dimensions of 512 × 512 × 3 undergoes sequential convolutional and max-pooling operations to generate five preliminary feature maps. Specifically, the process initiates with two 3 × 3 convolutional layers (kernel size: 3 × 3 and stride.

1) [1)]Producing 64-channel outputs, forming the first preliminary feature map. Subsequent stages involve a max-pooling operation (kernel size: 2 × 2 and stride of 2) followed by two 128-channel 3 × 3 convolutions for the second feature map.
2) Another max-pooling layer preceding three 256-channel 3 × 3 convolutions to derive the third feature map.
3) Repeated max-pooling combined with three 512-channel 3 × 3 convolutions for both the fourth and fifth feature maps.

Throughout this hierarchical structure, each max-pooling layer systematically reduces spatial dimensions by half while preserving channel depth, establishing a pyramidal feature representation that preserves multiscale contextual information for enhanced feature fusion in subsequent network modules.

The feature enhancement network employs a multistage upsampling and fusion strategy to integrate five preliminary effective feature layers. Specifically, the fifth preliminary feature layer undergoes upsampling and concatenation with the fourth layer, followed by dual convolutional operations to generate the first enhanced feature layer. This enhanced layer is subsequently upsampled and fused with the third preliminary layer through concatenation and dual convolutions to produce the second enhanced feature layer. The process iterates as the second enhanced layer is upsampled, concatenated with the second preliminary layer, and processed by dual convolutions to form the third enhanced feature layer. Finally, this third enhanced layer undergoes upsampling, merges with the first preliminary feature layer via concatenation, and completes dual convolutional operations to yield the ultimate target fusion feature layer, which maintains identical spatial dimensions as the original input image while containing 64 channels.

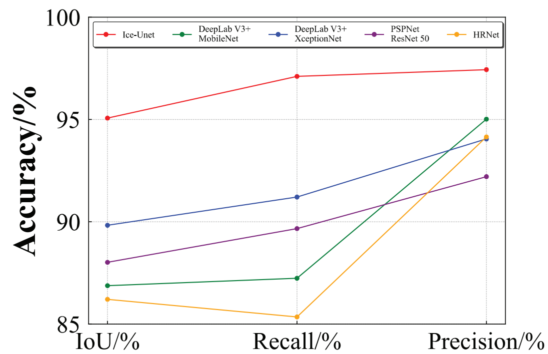

Fig. 3. Evaluation of different methods.

The third component, the prediction network, employs a 1 × 1 convolutional layer to adjust the channel dimensions of the final effective feature map, aligning it with the number of semantic categories. This operation transforms high-dimensional features into a task-specific semantic space, enabling pixel-wise semantic segmentation of sea ice surface features. By performing pointwise classification on each spatial location within the feature map, the network achieves precise delineation of ice-covered regions at the pixel level, effectively distinguishing between sea ice and background elements in remote sensing imagery. The model outputs pixel-wise classifications for four target classes: sea ice (red), melt ponds (green), open water (blue), and ice breaker (black).

D. *Training and Evaluation*

During the training step, approximately 7500 high-resolution images were randomly selected and annotated with pixel-level labels using ISAT-SAM [7]. This tool leverages

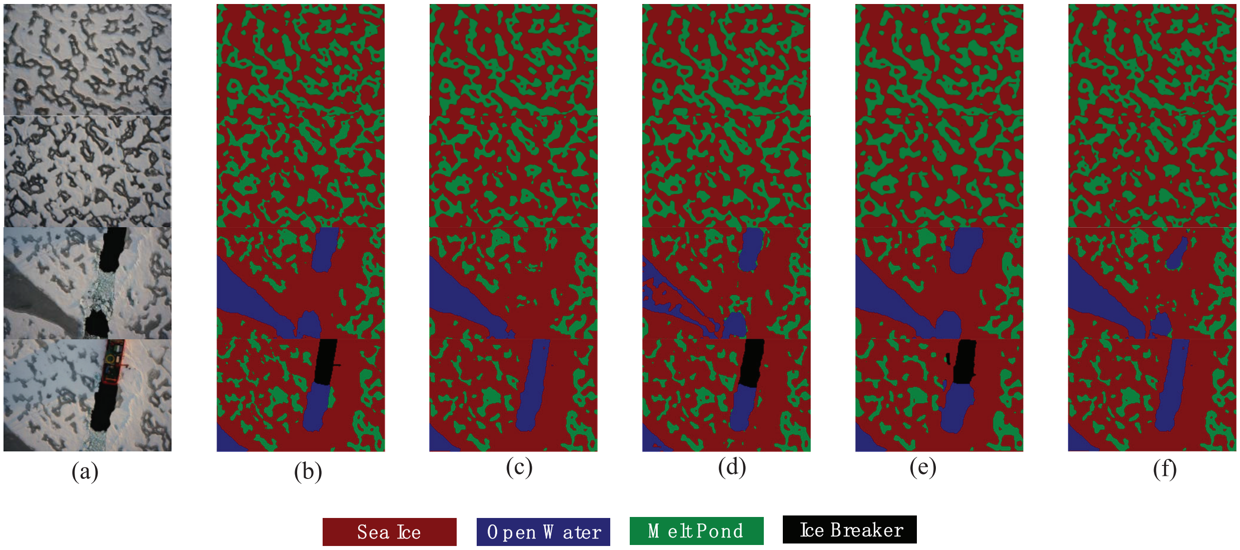

Fig. 4. Sea ice classification results using different methods. (a) UAV data. (b) Ice-Unet. (c) DeepLabV3+ (MobileNet). (d) DeepLabV3+ (Xception). (e) PSPNet (ResNet50). (f) HRNet.

SAM's advanced segmentation capabilities and interactive annotation features to efficiently generate precise labels for four classes: sea ice, melt ponds, open water, and ship. To optimize compatibility with deep learning-based sea ice classification models, the final dataset was partitioned into training (80%), validation (10%), and testing (10%) subsets to ensure robust model evaluation.

Acquiring large annotated datasets for sea ice segmentation remains challenging due to labor-intensive manual labeling and high computational demands. To address this, we implement transfer learning by initializing Ice-Unet with PASCAL VOC2012 pretrained weights, leveraging deep convolutional networks' feature extraction capabilities. This transfers generic visual recognition to sea ice analysis, enabling effective adaptation through fine-tuning on limited ice/melt pond datasets while reducing overfitting risks. Prior studies show such approaches achieve >98% classification accuracy with reduced parameters, particularly using modified U-Nets optimized for polar features [8], [9], [10], [11], [12].

Evaluation employs recall for target-class coverage, precision for prediction reliability, and IoU for spatial consistency between predictions and ground truth. Three metrics that evaluate semantic segmentation are given as follows:

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}} \tag{8}$$

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}} \tag{9}$$

$$\text{IoU} = \frac{\text{TP}}{\text{TP} + \text{FP} + \text{FN}} \tag{10}$$

where TP, FN, and FP represent the true positive, false negative, and false positive, respectively.

IV. Results and Analysis

A. Sea Ice Classification Using Different Methods

According to the experimental results presented in Table I and Figs. 3 and 4, the Ice-Unet model demonstrates optimal performance in both IoU and recall metrics, exhibiting superior consistency and stability compared to other architectures. The DeepLab V3+ (MobileNet) variant fails to recognize ship targets completely, resulting in notably inferior IoU and recall scores, while DeepLab V3+ (Xception) achieves relatively balanced performance across categories yet remains consistently outperformed by Ice-Unet in all metrics. Although PSPNet shows comparable overall performance to DeepLab V3+ (Xception), its classification accuracy for sea ice is significantly lower than that of DeepLab V3+ (Xception). For melt ponds, their intricate textural complexity and visual similarity to surrounding features constrain effectiveness. Besides, the limited representation of ice breaker in training data (less than 1.5%) results in insufficient feature learning, leading to reduced segmentation accuracy. Notably, HRNet exhibits the poorest recognition precision for sea ice leads despite achieving 100% precision in ship detection, suggesting potential overfitting tendencies in specific categories. While Ice-Unet attains the highest average precision score, DeepLab V3+ (Xception) demonstrates exceptional precision specifically for sea ice leads, highlighting the complementary strengths of different architectures in handling distinct semantic classes. This comprehensive evaluation underscores the critical importance of architectural selection based on target application requirements and class-specific performance characteristics.

B. MPF Estimation Based on Ice-Unet

Fig. 5 presents the sea ice classification and melt pond fraction (MPF) estimation results at the Test-01 site on August 26, 2024 [13], [14]. As shown in Fig. 5(a), extensive melt pond formation is observed across the study area. Notably, a continuous open water zone formed along the navigation track of the "Sun Yat-sen University Polar Research Vessel" (left portion of the image), while the trailing area exhibits partial refreezing characteristics. The classification results in Fig. 5(b) reveal substantial melt pond distribution with interconnected percolation features forming linear and reticulated leads. Specifically, the top-right region displays linear leads exceeding 400 m in length with 13 m width, while

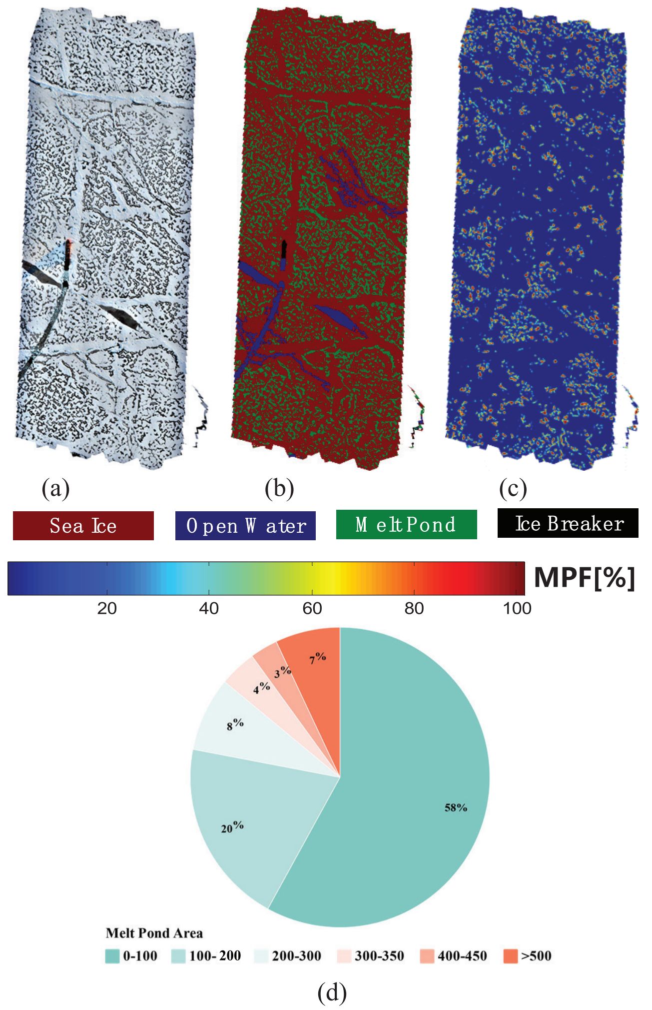

Fig. 5. Melt pond derived from Ice-Unet model. (a) UAV data. (b) Ice-Unet. (c) MPF. (d) Statistics of Melt Pond Area [m^2] (visualized using the chiplot: https://www.chiplot.online/).

the bottom-left section exhibits a network of interconnected channels where the longest lead measures approximately 370 m in length and 8 m in width. Spatial analysis in Fig. 5(c) and (d) indicates a homogeneous distribution of melt ponds, accounting for 25% of the total imaged area. Over 90% of the melt ponds are smaller than 500 m^2. Approximately 60% of these melt ponds occupy areas smaller than 100 m^2, with observed basal percolation and partial interconnections among adjacent ponds. These morphological characteristics suggest high developmental potential under continued melting and water infiltration processes, likely evolving into more extensive linear and reticulated lead systems. The observed spatial patterns align with recent findings on melt pond evolution dynamics, particularly regarding the transition threshold from discrete ponds to interconnected hydrological networks. The dimensional parameters of leads demonstrate scale-dependent morphological development consistent with fractal growth patterns observed in Arctic marginal ice zones.

V. Conclusion

This study proposes a novel approach for polar sea ice classification by leveraging UAV-based optical imagery and a deep learning framework, specifically the Ice-Unet model, to achieve high-precision semantic segmentation of sea ice. Utilizing high-resolution UAV images from the 2024 “Sun Yat-sen University Polar” Arctic expedition, the method involves geometric correction, cropping, and annotation via the ISAT-SAM tool to construct a dataset, enhanced by transfer learning for model optimization. The experimental results demonstrate that Ice-Unet outperforms mainstream architectures such as DeepLabV3+ (Xception/MobileNet), PSPNet, and HRNet in IoU, recall, and precision rates, achieving a mean accuracy of 97.44%, with exceptional performance in melt pond detection (IoU 91.46%) and ice lead characterization (IoU 94.48%). Case analysis reveals the model’s capability to capture the spatial distributions of melt pond and track the evolution of sea ice lead. This work validates the efficacy of Ice-Unet-based deep learning algorithm for dynamic sea ice monitoring, offering a high-resolution solution for sea ice classification and ship navigation.

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